THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

May 14, 2024 9:30 am – 12:00 noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the answer book.

Answer ALL SIX questions.

Notes:

- Write down your university number, seat number and model of calculator used on the cover of the answer book.
- The blank pages in the answer book can be used as additional answer space provided that you make a clear indication in the original answer space.
- Unless otherwise specified, you should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (27 marks) For this question, give answers only. No explanation is needed. Each question carries 3 marks.

(a) Let $f(x, y) = x\sqrt{y}$. Find the second Taylor polynomial of f at the point $(0, 1)$.

(b) Let T be the triangle (including the boundary and interior) with vertices $(-1, 0)$, $(1, 0)$ and $(0, 2)$. Rewrite the double integral

$$\iint_T f(x, y)\, dA$$

into the form

$$\int_{\square}^{\square} \int_{\square}^{\square} f(x, y)\, dx\, dy \quad \text{or} \quad \int_{\square}^{\square} \int_{\square}^{\square} f(x, y)\, dy\, dx,$$

where the boxes should contain appropriate lower and upper limits.

(c) Write out the double integral which can be used to find the area inside the cardioid $r = 3 - 2\cos\theta$ and outside the circle $r = 2$. (Do not evaluate the integral.)

(d) Write out the triple integral in spherical coordinates which can be used to find the volume of the solid that lies above the cone $z = \sqrt{3(x^2 + y^2)}$ and below $x^2 + y^2 + z^2 = 2z$. (Your answer should be a triple integral involving ρ, ϕ and θ. Do not evaluate the integral.)

(e) Find all local maximum points of the function $f(x, y) = 6 + x^3 + y^3 + xy$.

(f) Using ONE iteration of Newton-Raphson's method with initial guess $(1, -1)$ to find an approximation to the critical point of $f(x, y) = x^4 + y^2 + xy + 2y$. (Round your answer to 2 decimal places.)

Question (g)-(i). Let $A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 8 & 8 & 7 & 5 \end{bmatrix}$.

(g) Find the reduced row echelon form of A.

(h) Find the determinant of A.

(i) It is known that 10 is an eigenvalue of A. Find the corresponding eigenvector. (Let the first non-zero entry be 1.)

2. (24 marks) In each multiple choice question below, some numbered choices are given and some are correct (wordings like *is* and *are* do <u>not</u> indicate the singularity/plurality of the number of correct choices). Answer the question by adding up the numbers of the correct choices. In case all choices are wrong, answer 0. No explanation is needed. Each question carries 3 marks.

EXAMPLE: Which of the following is correct?

(1) $1 + 2 = 3$.

(2) $1 + 1 = 2$.

(4) $1/0 = 1$.

(8) $1 - 1 = 0$. Answer: 11.

(a) A point P has spherical coordinates $\left(2, -\frac{\pi}{3}, -\frac{\pi}{3}\right)$. When expressing P in rectangular or cylindrical coordinates, which of the following will be positive?

(1) x

(2) y

(4) z

(8) r

(b) Let A be a square matrix with reduced row echelon form R. Which of the following is true?

(1) If the rows of A are linearly independent, then the columns of A are linearly independent.

(2) If the rows of A are linearly independent, then the rows of R are linearly independent.

(4) If the first three rows of A are linearly independent, then the first three rows of R are linearly independent.

(8) If the first three columns of A are linearly independent, then the first three columns of R are linearly independent.

(c) Which of the following can be used to find the mass of a circular cone of base radius $r = 1$ and height $h = 4$, given that the density equals the distance from the vertex.

(1) $\displaystyle\int_0^{2\pi}\int_0^1\int_{4r}^4 r\sqrt{z^2+r^2}\,dz\,dr\,d\theta$

(2) $\displaystyle\int_0^{2\pi}\int_0^1\int_0^{4-4r} \sqrt{5}r^2\,dz\,dr\,d\theta$

(4) $\displaystyle\int_0^{2\pi}\int_0^1\int_0^{4-4r} r\sqrt{(4-z)^2+r^2}\,dz\,dr\,d\theta$

(8) $\displaystyle\int_0^{2\pi}\int_0^{\tan^{-1}\left(\frac{1}{4}\right)}\int_0^{4\sec\phi} \rho^3\sin\phi\,d\rho\,d\phi\,d\theta$

(d) Which of the following are saddle points of $f(x, y) = xy(2 - x - y)$?

(1) $(0, 0)$

(2) $(0, 2)$

(4) $(2, 0)$

(8) $\left(\frac{2}{3}, \frac{2}{3}\right)$

(e) An $n \times n$ matrix A is invertible if

(1) $\det A < 0$.

(2) A is row equivalent to I.

(4) $A\mathbf{x} = \mathbf{0}$ has non-trivial solutions.

(8) all eigenvalues of A are positive.

(f) Let A be an $m \times n$ matrix. If the system $A\mathbf{x} = \mathbf{b}$ has a unique solution, which of the following is possible?

(1) $m < n$ and $\mathbf{b} = \mathbf{0}$

(2) $m < n$ and $\mathbf{b} \neq \mathbf{0}$

(4) $m > n$ and $\mathbf{b} = \mathbf{0}$

(8) $m > n$ and $\mathbf{b} \neq \mathbf{0}$

(g) Let A be matrix, $\mathbf{v}$ be a column vector, and I (identity matrix) and O (zero matrix) have appropriate sizes so that all of the following multiplications are well-defined. Which of the following must be true?

(1) $O\mathbf{v} = 0\mathbf{v}$

(2) $I\mathbf{v} = 1\mathbf{v}$

(4) $AI = IA$

(8) $AO = OA$

(h) Let A and B be $n \times n$ matrices with real entries. Which of the following must be true?

(1) $\det(A^2) \geq 0$

(2) $\det(A^T) = \det A$

(4) $\det(AB) = \det(BA)$

(8) $\det(AB) = \det A \det B$

3. (18 marks) For each of the following statements, circle **T** if it is true and **F** if it is false, and then give a very brief explanation. Each question carries 3 marks.

(a) $\begin{bmatrix} 1 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ is a reduced row echelon matrix.

(b) Let rref M denote the reduced row echelon form of a matrix M. If A and B are square matrices of the same size and A is invertible, then rref $AB = (\text{rref } A)(\text{rref } B)$.

(c) A vector in a linearly dependent set must be a linear combination of the other vectors.

(d) If (a, b) is a critical point of $z = f(x, y)$, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

(e) The function

$$f(x, y) = \begin{cases} \dfrac{(x+y)^2}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$.

(f) Let A be an $m \times n$ matrix. If there exists a matrix B such that $BA = I$, then the system $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$.

4. (10 marks) Let $A = \begin{bmatrix} 4 & 8 & -2 \\ -3 & -6 & 1 \\ 9 & 12 & -5 \end{bmatrix}$. Determine whether A is diagonalizable. If yes, diagonalize it and find the $(3,3)$-entry of A^{2024}; if not, state the reason.

5. (11 marks) What is the minimum altitude that a mountain climber at the summit of a mountain can descend to, given that the altitude of the mountain is determined by the function $h(x,y) = 3000 - \frac{1}{10000}(5x^2 + 4xy + 2y^2)$ in meters, where x and y are horizontal coordinates on the earth, and the climber can reach anywhere in the region $x^2 + y^2 \leq 1000000$?

6. (10 marks) Let $\mathbf{x}$, $\mathbf{y}$ be linearly independent vectors in $\mathbb{R}^n$. Determine whether the vectors $\mathbf{u} = a\mathbf{x} + b\mathbf{y}$ and $\mathbf{v} = c\mathbf{x} + d\mathbf{y}$ are linearly dependent or independent, where a, b, c, d are real numbers.

******* **END OF PAPER** *******